

Tiago Davies

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WORK EXPERIENCE

Software Engineer

Aug 2025 – Present

Smart Software Solutions

- Implement secure **JWT bearer token authentication** with refresh token rotation for mobile clients in a cross-platform **.NET RESTful API**, configuring authentication middleware to preserve existing cookie-based sessions for the web platform.
- Design, implement, and maintain relational database schemas and business logic using **Entity Framework code-first migrations**, including a multi-allocation payment system, supporting partial payments against outstanding balances.
- Transform Figma designs into interactive and **responsive front-end views**, using modern CSS/SCSS and Bootstrap to achieve fluid UI across multiple screen sizes on both web and mobile.
- Modernize database transaction handling by introducing rollback/commit patterns and structured exception handling, to ensure **ACID compliance** and data integrity.
- Refactor **Angular front-end** architecture from template-driven to reactive forms, enforcing consistent client-side validation and error handling.
- Integrate FingerprintAIO biometric authentication on iOS and Android, enabling Face ID and fingerprint login with secure native credential handling.
- Integrate Tesseract Optical Character Recognition to extract and auto-fill fields with Machine Readable Zone data from passports, achieving over 90% parsing accuracy.
- Automate build deployments to the development server via bash scripts, reducing build times by 30%.

Computer Science Teaching Assistant

Jan 2023 – May 2024

University of Rochester

- Spearheaded weekly technical labs for 150+ students in core data structures and algorithms topics.
- Guided students through problem-solving, debugging, and code architecture during study sessions.

Computer Science Workshop Leader

Jan 2022 – Dec 2022

University of Rochester

- Led technical sessions for groups of 10+ students, engaged in collaborative discussion on algorithms.
- Participated in weekly leadership class to build actionable strategies for the upcoming workshops.

SKILLS

- **Languages:** C#, TypeScript, JavaScript, T-SQL
- **Back-end:** .NET 8/.NET (Core), Entity Framework, LINQ, ASP.NET Web API
- **Front-end:** Angular 17, React, HTML, CSS/SCSS, Bootstrap, jQuery, RxJS
- **Database:** Microsoft SQL Server
- **Tools:** Git/GitHub, npm, Vite, Docker, Postman, VS Code, Visual Studio

PROJECTS

Entity Component System Web Application (Rogue Survivor)

[May 2026](#)

- Engineered a high performance **data-oriented** web app in TypeScript by leveraging an Entity Component System, optimizing for cache locality via flat TypedArrays to handle 2,000+ active entities.
- Automated the release life-cycle by implementing a **CI/CD pipeline** using GitHub Actions to push production builds and deployment to GitHub pages.
- Built a spatial hashing collision system to reduce collision queries from $O(N^2)$ to average linear time.

EDUCATION

University of Rochester

Graduated May 2025

Bachelor of Science, Computer Science

- Cumulative GPA: 3.81 (out of 4.0) | Water Polo Club Business Manager | 3x school hackathon winner